FS10

PCB Hardware Design Description

Factual design overview based on the current KiCad layout

Source	hardware/FS10--Hardware-KiCad/FS1--ATmega32U4.kicad_pcb
Controller	ATmega32U4 (U2)
Scope	Layout choices, interfaces, power path, copper-pour topology

Board finish - The assembled FS10 PCB uses a purple solder mask. This description records circuit and layout choices and does not report impedance, noise, EMC, or thermal performance measurements.

1. Board organization and interfaces

The board organizes external connectors at the perimeter, with the ATmega32U4 controller (U2), clock support, and small-signal components toward the central and upper areas. Service and user-facing connections remain accessible around the board edge.

- USB Type-B connector (USB1) provides the USB interface. D4 is the three-line USB protection device, and the USB D+ and D- connections are routed through named connector-side and device-side nets.
- The M12 / field-wire interface is presented through the nine-position connector row (CN1), with silkscreen color labels used by the assembly instructions.
- The TFT header (J12) carries the display SPI signals and power; the ICSP1 2 x 3 header provides MOSI, MISO, SCK, RESET, +5 V, and ground for programming.
- U3 is the MAX232E RS-232 transceiver between the ATmega32U4 UART and the external FS10 interface.

2. Power path and buck-converter placement

The 24 V input enters through J1 on the lower board edge. The layout records a raw-input protection stage, an on-board power switch, and the AP63200WU-7 synchronous buck regulator (IC2). The buck power components are grouped near the lower-right edge: IC2, L1, its input/output capacitors, feedback components, and the +5 V isolation diode D1.

- D2 is recorded as a 28 V standoff TVS on the 24 V raw input; the switch separates the raw and switched 24 V nets.

- IC2 is specified for a 3.8 V to 32 V input range. L1 is a shielded 10 uH inductor; D1 isolates the buck output into the +5 V rail.
- The high-current conversion parts are grouped on the lower-right edge rather than dispersed around the microcontroller, display headers, and USB connector. This is a placement choice visible in the present layout, not a measured noise claim.

Layout record - The AP63200WU-7 buck stage follows the manufacturer-recommended local component arrangement: input/output capacitors, inductor, feedback network, and protection components are grouped with the converter at the power-entry area.

3. Copper pours and return-path topology

The current KiCad board contains a board-outline ground zone named GND on both F.Cu and B.Cu. The pour covers the board extent, with clearances and thermal connections defined in the source. Local +5 V copper zones and numerous connection-shaping zones are also present.

- The full-board GND zone provides a common reference copper area on both board sides.
- Dedicated local +5 V zones appear at multiple component connections; this is distinct from the board-wide GND pour.
- The source also contains many small zones used for pad/via connection shaping. They should not be described as analog isolation regions or impedance structures.

Copper-pour record - The GND zone is a continuous board-outline copper pour on both copper layers. This document does not characterize analog isolation, impedance, or noise performance.

4. Controller, display, sensing, and service choices

U2 is the ATmega32U4 in a TQFP-44 package. The board includes the X1 crystal network, UCAP support, AREF/AVCC support components, a reset switch, the USB data connection, and the headers used by the display, encoder, and programming fixture.

- The display connector J12 uses the SPI signals MOSI, MISO, SCK, TFT_CS, and TFT_DC, plus reset, +5 V, and ground.
- The ICSP1 connector exposes the standard AVR programming signals directly for AVRISP mkII programming.
- J11 is the 1 x 5 encoder-module header on the lower portion of the left board edge. The board silkscreen identifies this location as ENCODER.
- J13 is the optional 1 x 4 I2C/OLED header, immediately above J11 on the left edge. Its silkscreen identifies it as optional OLED I2C; the header is recorded as optional/DNP in the layout.
- SW2 is the 12 mm panel-button footprint on the right side of the board, adjacent to the silkscreen switch label. It is routed to the PANEL_SW_D7 signal.
- Field / relay status signals terminate at the controller inputs through the external harness connector; firmware currently polls the PF0 and PF1 relay-status inputs.
- Test points and labelled headers support manufacturing inspection and service access.

The display backlight connection is reworked from VDD to an ATmega32U4 GPIO pin, providing a hardware path for a future PWM dimming function.

5. USB and 5 V rail design choices

USB1 connects through the USB connector-side nets to the D4 protection device and onward to the ATmega32U4 USB device nets. The VBUS path includes fuse F1, the VUSB net, and the D6 Schottky isolation diode into the +5 V system rail. R2 and R3 form the USB-present sensing path to the controller input.

USB record - The connector-side and device-side USB nets are separately named, with D4 protection located at the connector. No differential-pair impedance, length-matching, or high-speed-compliance claim is made here.

6. Engineering record boundary

The KiCad schematic and PCB source define net connectivity, component values, copper geometry, and fabrication output for the current revision. This document is a concise design description for review; it is not a performance-verification report.